

### Assignment-3

1. Write a program to implement (a) pointer to an object (b) this pointer. Practice both '.' (dot operator) and '→' (arrow operator).

Code:-

```
ques1.cpp
1 //Assignment- 3
2 //Question -1
3 #include <iostream>
4 using namespace std;
5 class Demo {
6     int x;
7 public:
8     void set(int x) {
9         this->x = x;    // using this pointer
10    }
11    void show() {
12        cout << "Value: " << x << endl;
13    }
14 };
15 int main() {
16     Demo d;
17     d.set(10);    // dot operator
18     d.show();
19     Demo *ptr = &d;
20     ptr->set(20); // arrow operator
21     ptr->show();
22     return 0;
23 }
```

Output:-

```
C:\Users\asus\Desktop\oops\  ×  +  v
Value: 10
Value: 20

-----
Process exited after 0.2613 seconds with return value 0
Press any key to continue . . .
```

2. Write a program to swap private values of two classes using a friend function.

Code:-

```

ques1.cpp  ques2.cpp
1  //Assignment- 3
2  //Question -2
3  #include <iostream>
4  using namespace std;
5  class B; // forward declaration
6  class A {
7      int x;
8  public:
9      A(int a) { x = a; }
10     friend void swap(A &, B &);
11     void show() { cout << "A: " << x << endl; }
12 };
13 class B {
14     int y;
15 public:
16     B(int b) { y = b; }
17     friend void swap(A &, B &);
18     void show() { cout << "B: " << y << endl; }
19 };
20 void swap(A &a, B &b) {
21     int temp = a.x;
22     a.x = b.y;
23     b.y = temp;
24 }
25 int main() {
26     A a(10);
27     B b(20);
28     swap(a, b);
29     a.show();
30     b.show();
31 }

```

Output:-

```

C:\Users\asus\Desktop\oops\  ×  +  ∨
A: 20
B: 10

-----
Process exited after 0.2916 seconds with return value 0
Press any key to continue . . .

```

3. Write a program to add data objects of two different classes using friend functions.

Code:-

```

ques1.cpp  ques2.cpp  ques3.cpp
1  //Assignment- 3
2  //Question -3
3  #include <iostream>
4  using namespace std;
5  class B;
6  class A {
7      int x;
8  public:
9      A(int a) { x = a; }
10     friend int add(A, B);
11 };
12 class B {
13     int y;
14 public:
15     B(int b) { y = b; }
16     friend int add(A, B);
17 };
18 int add(A a, B b) {
19     return a.x + b.y;
20 }
21 int main() {
22     A a(5);
23     B b(7);
24     cout << "Sum: " << add(a, b);
25 }

```

Output:-

```

C:\Users\asus\Desktop\oops\  x  +  v
Sum: 12
-----
Process exited after 0.2614 seconds with return value 0
Press any key to continue . . . |

```

4. Write a program to demonstrate the working of friend class.

Code:-

```

ques1.cpp  ques2.cpp  ques3.cpp  ques4.cpp
1  //Assignment- 3
2  //Question -4
3  #include <iostream>
4  using namespace std;
5  class A {
6      int x;
7  public:
8      A() { x = 100; }
9      friend class B;
10 };
11 class B {
12 public:
13     void show(A a) {
14         cout << "Accessing private data: " << a.x;
15     }
16 };
17 int main() {
18     A a;
19     B b;
20     b.show(a);
21 }

```

Output:-

```

C:\Users\asus\Desktop\oops\  x  +  v
Accessing private data: 100
-----
Process exited after 0.352 seconds with return value 0
Press any key to continue . . .

```

5. Write a program using Array of Objects to display area of multiple rectangles.

Code:-

```

ques1.cpp  ques2.cpp  ques3.cpp  ques4.cpp  ques5.cpp
1  //Assignment- 3
2  //Question -5
3  #include <iostream>
4  using namespace std;
5  class Rectangle {
6      int l, b;
7  public:
8      void setData(int x, int y) {
9          l = x;
10         b = y;
11     }
12     void area() {
13         cout << "Area: " << l * b << endl;
14     }
15 };
16 int main() {
17     Rectangle r[3];
18     r[0].setData(2,3);
19     r[1].setData(4,5);
20     r[2].setData(6,7);
21     for(int i=0;i<3;i++)
22         r[i].area();
23 }

```

Output:-

```

C:\Users\asus\Desktop\oops\  x  +  v
Area: 6
Area: 20
Area: 42
-----
Process exited after 0.275 seconds with return value 0
Press any key to continue . . . |

```

6. Write a program to define function **cube()** as inline for calculating cube of a number.

Code:-

```

ques1.cpp ques2.cpp ques3.cpp ques4.cpp ques5.cpp ques6.cpp
1 //Assignment- 3
2 //Question -6
3 #include <iostream>
4 using namespace std;
5 inline int cube(int x) {
6     return x * x * x;
7 }
8 int main() {
9     cout << "Cube: " << cube(3);
10 }

```

Output:-

```

C:\Users\asus\Desktop\oops\ x + v
Cube: 27
-----
Process exited after 0.2347 seconds with return value 0
Press any key to continue . . .

```

7. Write a program to pass an object as an argument and return the object from a function.

- (a) Use pass-by-value
- (b) Use pass-by reference

Code:-(a)

```

ques1.cpp ques2.cpp ques3.cpp ques4.cpp ques5.cpp ques6.cpp ques7a.cpp
1 //Assignment- 3
2 //Question -7(a)
3 #include <iostream>
4 using namespace std;
5 class Test {
6     int x;
7 public:
8     Test(int a=0) { x = a; }
9     Test add(Test t) {
10         return Test(x + t.x);
11     }
12     void show() {
13         cout << x << endl;
14     }
15 };
16 int main() {
17     Test t1(10), t2(20), t3;
18     t3 = t1.add(t2);
19     t3.show();
20 }

```

Output:-

```

C:\Users\asus\Desktop\oops\  X  +  v
30
-----
Process exited after 0.3017 seconds with return value 0
Press any key to continue . . .

```

Code:-(b)

```

ques1.cpp  ques2.cpp  ques3.cpp  ques4.cpp  ques5.cpp  ques6.cpp
1  //Assignment- 3
2  //Question -7b
3  #include <iostream>
4  using namespace std;
5  class Test {
6      int x;
7  public:
8      Test(int a=0) { x = a; }
9      Test add(Test &t) {
10         return Test(x + t.x);
11     }
12     void show() {
13         cout << x << endl;
14     }
15 };
16 int main() {
17     Test t1(15), t2(25), t3;
18
19     t3 = t1.add(t2);
20     t3.show();
21 }

```

Output:-

```

C:\Users\asus\Desktop\oops\  X  +  v
40
-----
Process exited after 0.2382 seconds with return value 0
Press any key to continue . . .

```